2 **Poisson & binomial distributions**	2.1	**The Poisson distribution** **The additive property of Poisson distributions**	**Students will be expected to use this distribution to model a real-world situation and to comment critically on the appropriateness.** **Students will be expected to use their calculators to calculate probabilities including cumulative probabilities.** **Students will be expected to use the additive property of the Poisson distribution. E.g. if X = the number of events per minute and $X \sim \mathrm{Po}(\lambda)$, then the number of events per 5 minutes ~ $\mathrm{Po}(5\lambda)$.** **If X and Y are independent random variables with $X \sim \mathrm{Po}(\lambda)$ and $Y \sim \mathrm{Po}(\mu)$, then $X + Y \sim \mathrm{Po}(\lambda + \mu)$** **No proofs are required.**
	2.2	**The mean and variance of the binomial distribution and the Poisson distribution.**	**Knowledge and use of :** **If $X \sim \mathrm{B}(n, p)$, then $\mathrm{E}(X) = np$ and $\mathrm{Var}(X) = np(1 - p)$** **If $Y \sim \mathrm{Po}(\lambda)$, then $\mathrm{E}(Y) = \lambda$ and** **$\mathrm{Var}(Y) = \lambda$** **Derivations are not required.**
	2.3	**The use of the Poisson distribution as an approximation to the binomial distribution.**	**When n is large and p is small the distribution $\mathrm{B}(n, p)$ can be approximated by $\mathrm{Po}(np)$.** **Derivations are not required.**